MICROCONTROLLER PROJECT

(PCCST504)

PLANT WATERING REMINDER USING STM32 AND SOIL MOISTURE SENSOR

SUBMITTED TO

Ms.ARYALEKSHMI A

SUBMITTED BY

SBC24CS061	ANJANA R
SBC24CS073	ARJUN PRASAD
SBC24CS134	KASINATH R KRISHNA
SBC24CS156	NEERAJA RAJESH

ABSTRACT

The Plant Watering Reminder using STM32 and Soil Moisture Sensor is a simple embedded system designed to monitor the moisture level of soil and remind the user when a plant needs watering. The system uses a soil moisture sensor to detect the amount of moisture present in the soil. The sensor output is connected to the ADC of an STM32 microcontroller, which reads and processes the moisture level. When the soil becomes dry and the moisture level falls below a predefined threshold, the STM32 activates a buzzer or LED to provide a watering reminder. When sufficient moisture is available, the indicator remains OFF. The project uses an ARM Cortex-M based STM32, ADC, GPIO, Embedded C, and basic delay/timer concepts. This system helps prevent plants from being under-watered and provides a simple and convenient method for monitoring soil moisture. The project is low-cost, easy to implement, and demonstrates basic sensor interfacing and microcontroller programming.